

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273063

Kind Code

A1

Publication Date

August 28, 2025

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ALERT SYSTEM AND METHOD THEREOF

Abstract

An alert system for a vehicle includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that, when executed on the data processing hardware, cause the data processing hardware to perform operations. The operations include receiving, from a user device, location data, determining, based on the location data, a position of the user device relative to the vehicle, and configuring an electronic control unit (ECU) with a range. The operations also include issuing, based on the position of the user device and the range, a vibration request and executing, based on the issued vibration request, a user alert application of the ECU.

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Appl. No.: 18/586503

Filed: February 25, 2024

Publication Classification

Int. Cl.: G08B21/18 (20060101)

U.S. Cl.:

CPC G08B21/182 (20130101);

Background/Summary

INTRODUCTION

[0001] The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0002] The present disclosure relates generally to an alert system for a vehicle to support people with hearing assistance devices or distracted users by using vibration on a user device.

[0003] Vehicles are typically equipped with audible alerts configured to notify a user of an event related to the vehicle. For example, if a user is not wearing a seatbelt, the vehicle may issue a chime or other audible alert until the seatbelt is fastened. While an audible alert is effective at reminding users of different functions of or related events to the vehicle, persons with hearing impairments may benefit from additional forms of alerts. Vehicles may also prompt, in combination with the audible alert, a text notification on a user interface of the vehicle. While a text notification may be advantageous to a person with hearing impairments, the person may overlook the text notification during the period of time the text notification is displayed. Thus, there is a need for improved alerts and notifications in relation to vehicle notifications.

SUMMARY

[0004] In some aspects, a computer-implemented method when executed by data processing hardware causes the data processing hardware to perform operations. The operations include determining, via location data, a location of a user device relative to a vehicle, identifying, based on positioning data and a vehicle status, a trigger event, and issuing, via a user alert application, a vibration request to the user device. The operations also include receiving, from the user device, a vibration protocol, determining, based on the trigger event and the vibration protocol, an alert level, and executing, based on the alert level, an alert on the user device.

[0005] In some examples, the alert level may include a low level, an intermediate level, and a high level, and the vibration protocol may include a single vibration, a multi-vibration, and an extended vibration. The single vibration may correspond to the low level, the multi-vibration may correspond to the intermediate level, and the extended vibration may correspond to the high level. The operations may include executing a calibration protocol and determining, based on the calibration protocol, the distance between a vehicle and the user device. In some instances, determining the location of the user device may include detecting a digital key and matching the detected digital key to a stored digital key profile. In other examples, determining the location of the user device may include determining distance data and identifying a change in the distance data corresponding to the user device. Optionally, the operations may include determining a range corresponding to the distance data of the user device. In some instances, determining the alert level may include comparing the change in the distance data with the determined range.

[0006] In other aspects, a system includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that, when executed on the data processing hardware, cause the data processing hardware to perform operations. The operations include determining, via location data, a location of a user device relative to a vehicle, identifying, based on positioning data and a vehicle status, a trigger event, and issuing, via a user alert application, a vibration request to the user device. The operations also include receiving, from the user device, a vibration protocol, determining, based on the trigger event and the vibration protocol, an alert level, and executing, based on the alert level, an alert on the user device.

[0007] In some examples, the alert level may include a low level, an intermediate level, and a high level, and the vibration protocol may include a single vibration, a multi-vibration, and an extended vibration. The single vibration may correspond to the low level, the multi-vibration may correspond to the intermediate level, and the extended vibration may correspond to the high level.

The operations may include executing a calibration protocol including detecting a digital key. In some instances, determining the location of the user device includes matching the detected digital key to a stored digital key profile. In other instances, determining the location of the user device includes determining distance data and identifying a change in the distance data corresponding to the user device. Optionally, the operations may include determining a range corresponding to the distance data of the user device. In further examples, determining the alert level includes comparing the change in the distance data with the determined range.

[0008] In further aspects, an alert system for a vehicle includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that, when executed on the data processing hardware, cause the data processing hardware to perform operations. The operations include receiving, from a user device, location data, determining, based on the location data, a position of the user device relative to the vehicle, and configuring an electronic control unit (ECU) with a range. The operations also include issuing, based on the position of the user device and the range, a vibration request and executing, based on the issued vibration request, a user alert application of the ECU.

[0009] In some examples, the operations may include identifying, via the user alert application, a trigger event and issuing an alert, via the user alert application, in response to the trigger event. Optionally, the operations may include comparing the position of the user device with the configured range, and executing the user alert application includes issuing an alert in response to the comparison of the position of the user device with the configured range. In some instances, the operations may include executing, in response to the issued vibration request, a vibration protocol, the vibration protocol corresponding to the alert.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The drawings described herein are for illustrative purposes only of selected configurations and are not intended to limit the scope of the present disclosure.

[0011] FIG. 1 is an example schematic of a vehicle with a user walking away from the vehicle with a user device equipped with an alert application according to the present disclosure;

[0012] FIG. 2 is an example schematic of a user walking away from a vehicle at different distances, the user wearing a user device equipped with an alert application according to the present disclosure;

[0013] FIG. 3 is an example block diagram of an alert system according to the present disclosure;

[0014] FIG. 4 is a partial perspective view of an interior of a vehicle equipped with an alert system according to the present disclosure;

[0015] FIG. 5 is an example flow diagram for an alert system according to the present disclosure; and

[0016] FIG. 6 is another example flow diagram for the alert system of FIG. 5.

[0017] Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0018] Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

[0019] The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

[0020] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” “attached to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” “directly attached to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0021] The terms “first,” “second,” “third,” etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

[0022] In this application, including the definitions below, the term “module” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor (shared, dedicated, or group) that executes code; memory (shared, dedicated, or group) that stores code executed by a processor; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

[0023] The term “code,” as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, and/or objects. The term “shared processor” encompasses a single processor that executes some or all code from multiple modules. The term “group processor” encompasses a processor that, in combination with additional processors, executes some or all code from one or more modules. The term “shared memory” encompasses a single memory that stores some or all code from multiple modules. The term “group memory” encompasses a memory that, in combination with additional memories, stores some or all code from one or more modules. The term “memory” may be a subset of the term “computer-readable medium.” The term “computer-readable medium” does not encompass transitory electrical and electromagnetic signals propagating through a medium, and may therefore be considered tangible and non-transitory memory. Non-limiting examples of a non-transitory memory include a tangible computer readable medium including a nonvolatile memory, magnetic storage, and optical storage.

[0024] The apparatuses and methods described in this application may be partially or fully implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on at least one non-

transitory tangible computer readable medium. The computer programs may also include and/or rely on stored data.

[0025] A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

[0026] The non-transitory memory may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by a computing device. The non-transitory memory may be volatile and/or non-volatile addressable semiconductor memory. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0027] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0028] Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICS (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0029] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile

memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0030] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0031] Referring to FIGS. 1-3, an alert system **10** for a vehicle **100** includes an electronic control unit (ECU) **12** of the vehicle and a user device **200** in communication with the ECU **12**. The ECU **12** is configured with a user alert application **14**, which is executed by data processing hardware **16** of the ECU **12**. The ECU **12** also includes memory hardware **18** in communication with the data processing hardware **16**. The memory hardware **18** stores instructions that, when executed on the data processing hardware **16**, cause the data processing hardware **16**, to perform operations described herein. For example, the data processing hardware **16** is configured to execute the user alert application **14** to issue alerts **20** from the user alert application **14** to the user device **200**. The alert system **10** is configured to advantageously provide tactile feedback to a user that may or may not utilize hearing assist devices based on a hearing status of the user. Thus, the alert system **10** advantageously provides tactile feedback for the user regardless of the hearing status of the user.

[0032] The user alert application **14** is configured to issue the alert **20** or alerts **20** with varying degrees of alert levels **22**, such that the user device **200** receives the alert level **22** to support, for example, a hearing amplification device (not shown). For example, the alert levels **22** may include, but are not limited to, a low level **22a**, an intermediate level **22b**, and a high level **22c**. The alert levels **22** may also have a variable range that may capture levels between each of the low, intermediate, and high levels **22a-22c**. For example, the alert levels **22** may progressively increase between the different alert levels **22**. Each of the alert levels **22** are described in more detail below with respect to the vehicle **100** and the user device **200** and operation of the alert system **10**. The ECU **12** is equipped with various short-range wireless communication protocols including, but not limited to, Bluetooth® Low Energy (BLE), and ultra-wide band (UWB). For example, the ECU **12** may detect, via UWB, when a user **210** is positioned within the vehicle **100** and, when the user **210** exits the vehicle **100**, may utilize BLE to determine positioning data **24** based on BLE communication with the user device **200**.

[0033] Further, the ECU **12** may utilize BLE channel sounding (BLE/CS) to measure a distance D.sub.1-D.sub.n between the user device **200** and the vehicle **100**. Channel sounding (CS) is a subset of Bluetooth® technology that measures a distance between a user device **200** and the equipped CS device (e.g., the ECU **12**) by calculating a series of frequencies to estimate a distance D.sub.1-D.sub.n between the user device **200** and the ECU **12**. Thus, the ECU **12** may utilize CS to estimate location data **202** and, more specifically, distance data **202a** of the user device **200**. For example, the ECU **12** may use UWB and BLE/CS as a complement in function of a range of each of the UWB and BLE/CS. The distance data **202a** is utilized by the user alert application **14** in determining whether to issue an alert **20** and the alert level **22**, as described herein. The user alert application **14** utilizes BLE/CS to determine the position of the user **210** and subsequently

determine an order of issuing the alerts **20**. For example, the user alert application **14** may determine, based on the distance data **202a**, a hierarchy of the alert levels **22**, such that the alert **20** having the highest priority alert level **22** may be issued at an earlier distance D.sub.1-D.sub.n. [0034] In other examples, the ECU **12** may utilize a BLE received signal strength indicator (RSSI), which evaluates the power of the transmission between the user device **200** and the ECU **12**. For example, the ECU **12** may utilize a combination of UWB and BLE RSSI in determining a signal strength of the user device **200** and, thus, to estimate the distance data **202a** by utilizing a complement in function of a range of each of the UWB and BLE RSSI, respectively. In yet another example, the ECU **12** may utilize a combination of UWB and Wi-Fi® round trip wave time to estimate the distance data **202a**. In other examples, the ECU **12** may utilize a combination of UWB, BLE RSSI, and CS and/or a combination of UWB, BLE RSSI, CS, and Wi-Fi® round trip wave time to estimate the distance data **202a**. In these examples, the UWB provides a high accuracy estimation at a short range, while each of BLE/CS, BLE RSSI, and/or Wi-Fi® round trip wave time each provides an intermediate accuracy estimation at long range. Further, each of the above combinations of UWB, BLE/CS, BLE RSSI, and/or Wi-Fi® round trip wave time may be calibrated as part of the alert system **10**.

[0035] Referring still to FIGS. **1-3**, the memory hardware **18** may store a digital key profile **30** that corresponds to a digital key **204** of the user device **200**. For example, a user **210** pairs the user device **200** with the vehicle **100** via the ECU **12** by setting up the digital key profile **30**. In some examples, the user **210** may execute a digital key application on the user device **200** to pair the user device **200**, with the digital key **204**, to the ECU **12**. The user device **200** may include, but is not limited to, a cell phone, a tablet, and/or a smart watch. As depicted, the user device **200** is a smart watch worn by the user **210**. However, the user device **200** may be any device capable of receiving information or data from the ECU **12**.

[0036] Once paired, the user device **200** may send information to and receive information from the ECU **12**, including the alerts **20**. During the pairing process, the ECU **12** may execute a calibration protocol **26** to calibrate a range **32** corresponding to distances D.sub.1-D.sub.n of the user device **200** relative to the ECU **12** and, thus, the vehicle **100**. For example, the user alert application **14** may utilize the calibration protocol **26** to set alerts **20** at predetermined range(s) **32**. In some examples, the vehicle application **208** may communicate with the user alert application **14** that the user device **200** has moved a first distance D.sub.1 from the vehicle **100**, and the user alert application **14** may issue an alert **20** in response.

[0037] The distances D.sub.1-D.sub.n may include the first distance D.sub.1, a second distance D.sub.2, and a third distance D.sub.3. Similar to the alert levels **22**, the distances D.sub.1-D.sub.3 may span a varying degree of distances D.sub.1-D.sub.n within the predetermined range **32** stored in the memory hardware **18**. Thus, each distance D.sub.1-D.sub.3 may have a respective range **32** that may at least partially overlap with or extend from a previous distance D.sub.1-D.sub.3. With respect to the first distance D.sub.1, the range **32** is determined based on the ECU **12** losing UWB contact with the user device **200** and instead utilizing the BLE/CS to monitor the distance data **202a** of the user device **200**. The first distance D.sub.1 may overlap with the range **32** in which the ECU **12** may utilize UWB and BLE/CS to minimize a lost connection and maintain tracking of the distance data **202a**.

[0038] Referring to FIGS. **2-4**, the distances D.sub.1-D.sub.3 generally correspond to various trigger events **40** that may occur relative to the vehicle **100**. However, it is also contemplated that the trigger events **40** may occur within the vehicle when the ECU **12** is utilizing UWB to monitor the positioning data **24** and/or location data **202** of the user device **200**. FIGS. **2-4** generally depict examples of trigger events **40**. For example, the user **210** may enter the vehicle **100** and partially close a door **102** of the vehicle **100**. Thus, the partially closed door **102** may be a trigger event **40** of the user alert application **14**, and the user alert application **14** may issue an alert **20** in response. The user alert application **14** may elevate the alert level **22** based on the duration of the alert **20**.

For example, if the user **210** does not respond to the alert **20**, the user alert application **14** may elevate a low level alert **22a** to the intermediate level **22b**.

[0039] In other examples, lighting of the vehicle **100** may be operational, and the user **210** may exit the vehicle **100**. The user alert application **14**, in response to receiving the distance data **202a**, may issue a low level alert **22a** to remind the user **210** that the lighting has remained operational. The user may dismiss the alert **20** and/or may utilize the digital key **204** to cease operation of the lighting. In some instances, the user **210** may leave the digital key **204** or a key fob in the vehicle **100**. The user alert application **14** may determine the location data **202** of the user **210** based on the user device **200** and may issue an alert **20** in response. For example, the user may leave a mobile device within the vehicle **100**, and the user alert application **14** may issue the alert to a smart watch **200** worn by the user **210**. The ECU **12** may track, via the positioning data **24**, when the user **210** exits the vehicle **100** by determining that the user device **200** is outside of UWB range. If the user **210** leaves a device equipped with the digital key **204** in the vehicle **100**, the user alert application **14** may alert the worn user device **200** (e.g., a smart watch) based on the distance data **202a**.

[0040] When the user alert application **14** executes the calibration protocol **26**, the user alert application **14** calibrates, in addition to the range **32**, the alert level **22**. The user device **200** may also be configured with a vibration protocol **206**, which may be stored as part of a vehicle application **208** on the user device **200**. The vehicle application **208** may store and execute the digital key **204** as well as provide the connection with the ECU **12** to receive the location and distance data **202**, **202a**. The vibration protocol **206** may include, but is not limited to, a single vibration **206a**, a multi-vibration **206b**, and/or an extended vibration **206c**. The vibration protocol **206** may be shared with the user alert application **14** in response to a vibration request **28** issued by the user alert application **14** as part of the alert **20**. The vibration protocol **206** provides advantageous assistance to users that utilize hearing aids or other hearing amplification devices. Further, the vibration protocol **206** may assist users that may otherwise intentionally or unintentionally disregard audio or text alerts to further assist in alerting a user regardless of hearing status.

[0041] Upon receiving the vibration protocol **206**, the user alert application **14** may determine which vibration **206a-206c** of the vibration protocol **206** to execute based on the alert level **22**. In some examples, the user **210** may configure the vibration protocol **206** to correspond to particular alert levels **22**. For example, the single vibration **206a** may correspond to the low level **22a**, the multi-vibration **206b** may correspond to the intermediate level **22b**, and the extended vibration **206c** may correspond to the high level **22c**. However, various applications of the vibration protocol **206** may be utilized, such that each vibration **206a-206c** of the vibration protocol **206** may be interchangeably used with the various alert levels **22**. In other examples, the user alert application **14**, in response to receiving the vibration protocol **206**, may set the vibration **206a-206c** for each of the alert levels **22**.

[0042] The user alert application **14** cooperates with the vibration protocol **206** of the user device **200** to vibrate the user device **200** when the user alert application **14** issues an alert **20**. The user alert application **14** may determine which vibration **206a-206c** to associate with a respective alert level **22** and may communicate the determined vibration **206a-206c** with the user device **200**. The user device **200** may then execute the corresponding vibration protocol **206** to vibrate the user device **200** in accordance with the alert level **22**.

[0043] As mentioned above, the user alert application **14** issues the alert **20** in response to one or more trigger events **40**. The trigger event **40** may include, but is not limited to, a seat belt reminder, a door being ajar, lighting operations, a vehicle status, window status, roof status, and/or any other event that may trigger or result in a chime or alarm. The user alert application **14** utilizes the combination of the detection of the trigger event **40** and the distance data **202a** to determine the alert **20**. In some instances, the ECU **12** may determine a change in the distance data **202a** corresponding to the user device **200**. For example, the first distance D.sub.1 may correspond with

approximately ten (10) meters, the second distance D.sub.2 may correspond with approximately twenty (20) meters, and a third distance D.sub.3 may correspond with approximately thirty (30) meters. Each distance D.sub.1-D.sub.3 may be related to an alert level **22** and a specific trigger event **40** determined by the user alert application **14**.

[0044] In some examples, the first distance D.sub.1 may correspond to trigger events **40** having a high level **22c**. For example, if the user **210** leaves the engine operational, the user alert application **14** may issue an alert **20** with a high level **22c** corresponding to the extended vibration **206c**. Thus, the user alert application **14**, in response to determining the trigger event **40** and corresponding alert level **22**, issues the vibration request **28** to the user device **200**. The vibration request **28**, in this example, prompts the user device **200** to execute the extended vibration **206c** of the vibration protocol **206**.

[0045] In other examples, the second distance D.sub.2 may correspond to the door **102** being left ajar or a window **104** left partially or fully open, such that as the user approaches the second distance D.sub.2. the user alert application **14** identifies the trigger event **40**. In response, the user alert application **14** may issue the vibration request **28** to the user device **200** including the alert **20** with the intermediate level **22b**. In further examples, the third distance D.sub.3 may correspond to lighting remaining operational, as mentioned above. In this example, the user alert application **14** may issue the vibration request **28** including an alert **20** having a low level **22a**. The user alert application **14** may customize the vibration **206a-206c** based on the alert level **22** by communicating the alert level **22** with the user device **200**. Thus, the user alert application **14** may determine the alert level **22** by comparing a change in the distance data **202a** with the range **32**.

[0046] As mentioned above, the user alert application **14** utilizes BLE to communicate the vibration request **28** to the user device **200**. For example, the user alert application **14** issues the vibration request **28** using BLE to the vehicle application **208** on the user device **200** requesting execution of the vibration protocol **206**. When the user **210** is within the vehicle **100**, the ECU **12** may utilize UWB to identify the location data **202** of the user and, as the user **210** exits the vehicle **100**, may switch to BLE to monitor the distance data **202a** in addition to the location data **202**.

[0047] To determine that the user **210** is within the vehicle **100**, the ECU **12** may utilize a positioning method based on a signal arrival time to estimate the location data **202** of the user **210**. For example, the ECU **12** may utilize trilateral ranging using UWB. Once identified, the ECU **12** shares the location data **202** with the user alert application **14** and may issue an alert **20** depending on the position of the user **210** within the vehicle **100**. In determining whether the user has exited the vehicle **100**, the ECU **12** first may determine whether the user device **200** is outside of the UWB range. If the user device **200** is outside of the UWB range, then the ECU **12** may switch to using BLE/CS by utilizing a phase shift analysis of different tones between a received radio signal and a transmitted radio signal between the ECU **12** and the user device **200**.

[0048] Depending on the determined location data **202** and/or distance data **202a** calculated by the ECU **12**, the user may receive a distinguishable vibration **206a-206c** corresponding to the alert **20**. Thus. The alert system **10** utilizes a hybrid UWB and BLE/CS technique to identify the trigger event(s) **40** and subsequently notify the user via the user alert application **14**.

[0049] Referring now to FIGS. **5** and **6**, an example flow diagram for the alert system **10** is set forth. At step **400**, the user device **200** is paired with the ECU **12** of the vehicle **100**, and at step **402**, the vehicle **100** is connected to the user device **200**. The ECU **12** determines, at **404**, whether the user **210** is inside the vehicle **100** using a matrix of UWB devices spread out around the vehicle **100**. If the user **210** is not inside the vehicle **100**, the ECU **12** proceeds as described below with respect to steps **412-424**. If the user **210** is inside the vehicle **100**, then the ECU **12** determines, at **406**, whether a door **102** is ajar. If the door **102** is not ajar, then the ECU **12** determines, at **408**, whether a seatbelt chime is active. If the door **102** is not ajar and the seatbelt chime is not active, then the ECU **12** returns to monitor whether the user **210** is inside the vehicle **100**. If the door **102** is ajar and/or the seatbelt chime is active, then the ECU **12** executes the user alert application **14** to

send the vibration request **28** to the user device **200**. While the trigger events **40** described with respect to the flow diagram in FIG. **5** are a door **102** being ajar and/or a seatbelt chime, the trigger event **40** may be any of the trigger events described herein. and the flow diagram for the alert system **10** may proceed accordingly.

[0050] As mentioned above, if the user **210** is outside of the vehicle **100**, then the ECU **12** monitors, at **412**, a distance D.sub.1-D.sub.3 of the user **210** using BLE Channel Sounding. The ECU **12** determines, at **414**, whether the user **210** is approaching the first distance D.sub.1. If yes, then the ECU **12** may execute, at **416**, the user alert application **14** to issue an alert **20** with a low level **22a**. If the user **210** is past the first distance D.sub.1, then the ECU **12** determines, at **418**, whether the user **210** is approaching the second distance D.sub.2. If yes, then the ECU **12** may execute, at **420**, an alert **20** with the intermediate level **22b**. If the user **210** is past the second distance D.sub.2, then ECU **12** may determine, at **422**, whether the user **210** is approaching the third distance D.sub.3. If the user **210** is approaching the third distance D.sub.3, then the ECU **12** may execute, at **424**, the user alert application **14** to issue an alert **20** with a high level **22c**. If the user **210** is not approaching the third distance D.sub.3 or is past, the ECU **12** continues to monitor the distance data **202a**.

[0051] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

[0052] The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A computer-implemented method when executed by data processing hardware causes the data processing hardware to perform operations comprising: determining, via location data, a location of a user device relative to a vehicle; identifying, based on positioning data and a vehicle status, a trigger event; issuing, via a user alert application, a vibration request to the user device; receiving, from the user device, a vibration protocol; determining, based on the trigger event and the vibration protocol, an alert level; and executing, based on the alert level, an alert on the user device.
2. The method of claim 1, wherein the alert level includes a low level, an intermediate level, and a high level, and the vibration protocol includes a single vibration, a multi-vibration, and an extended vibration.
3. The method of claim 2, wherein the single vibration corresponds to the low level, the multi-vibration corresponds to the intermediate level, and the extended vibration corresponds to the high level.
4. The method of claim 1, further including executing a calibration protocol and determining, based on the calibration protocol, the distance between a vehicle and the user device.
5. The method of claim 1, wherein determining the location of the user device includes detecting a digital key and matching the detected digital key to a stored digital key profile.
6. The method of claim 1, wherein determining the location of the user device includes determining distance data and identifying a change in the distance data corresponding to the user device.
7. The method of claim 6, further including determining a range corresponding to the distance data of the user device.
8. The method of claim 7, wherein determining the alert level includes comparing the change in the

distance data with the determined range.

9. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: determining, via location data, a location of a user device relative to a vehicle; identifying, based on positioning data and a vehicle status, a trigger event; issuing, via a user alert application, a vibration request to the user device; receiving, from the user device, a vibration protocol; determining, based on the trigger event and the vibration protocol, an alert level; and executing, based on the alert level, an alert on the user device.

10. The system of claim 9, wherein the alert level includes a low level, an intermediate level, and a high level, and the vibration protocol includes a single vibration, a multi-vibration, and an extended vibration.

11. The system of claim 10, wherein the single vibration corresponds to the low level, the multi-vibration corresponds to the intermediate level, and the extended vibration corresponds to the high level.

12. The system of claim 9, further including executing a calibration protocol including detecting a digital key.

13. The system of claim 12, wherein determining the location of the user device includes matching the detected digital key to a stored digital key profile.

14. The system of claim 9, wherein determining the location of the user device includes determining distance data and identifying a change in the distance data corresponding to the user device.

15. The system of claim 14, further including determining a range corresponding to the distance data of the user device.

16. The system of claim 15, wherein determining the alert level includes comparing the change in the distance data with the determined range.

17. An alert system for a vehicle, the alert system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: receiving, from a user device, location data; determining, based on the location data, a position of the user device relative to the vehicle; configuring an electronic control unit (ECU) with a range; comparing the position of the user device with the configured range and issuing an alert in response to the comparison of the position of the user device with the configured range; issuing, based on the position of the user device and the range, a vibration request; and executing, based on the issued vibration request, a user alert application of the ECU.

18. The alert system of claim 17, further including identifying, via the user alert application, a trigger event and issuing an alert, via the user alert application, in response to the trigger event.

19. The alert system of claim 17, further including executing, in response to the issued vibration request, a vibration protocol, the vibration protocol corresponding to the alert.

20. A vehicle incorporating the alert system of claim 17.
